

Class Test 2

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Electricity and Electronics EBN122

3 October 2025

MEMO

Assessment Information

Maximum marks:	16	Full marks:	15
Duration of assessment:	30 min (excluding additional time)	Open/Closed book:	Closed
Additional time allocations:	5 min before and 5 min after for OCR answer sheet completion	Allowable materials:	Pencil case, non-programmable calculator
Extra-time venue:	Same as test venue	Submission format:	OCR
Total number of pages (this page included)		7	

Lecturers: Prof JP de Villiers, Prof JP Jacobs, Mr S Twala**Important**

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of OCR answer sheet 00/01.
4. Read the instructions on OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. Final answers must be written as real numbers and not as fractions.
7. Use **at least 3 significant digits** to write real numbers. This means that the real and imaginary parts of complex numbers should each be written using at least 3 significant digits, for example $Z = 10.0 + j0.283 \Omega$ (the same applies to magnitudes and angles).
8. Exact numbers in which decimal expansions are zero, can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

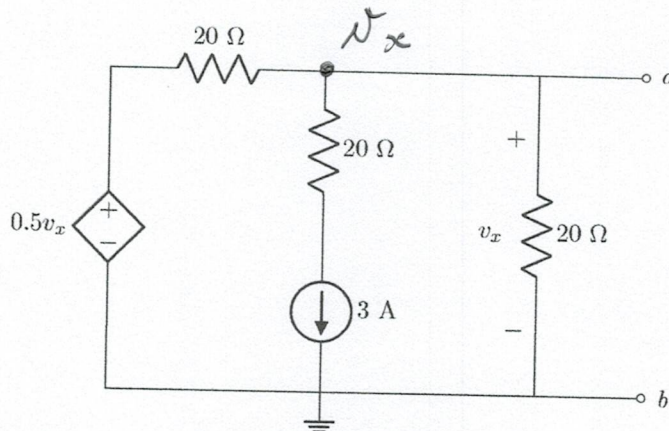
INSTRUCTIONS**Do step 0 in the first 5 minutes of the assessment. (-5 to 0 min)****Do steps 1 and 2 in the next 35 minutes of the assessment. (0-35 min)****TIME: -5 min to 0 min****STEP 0 :** Complete the front page of OCR answer sheet 00/01, and also fill in your student number on additional OCR answer sheets.**TIME: 0 min to 35 min****STEP 1:** Do your calculations on the question paper.**STEP 2:** Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN122C02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 2 [4]

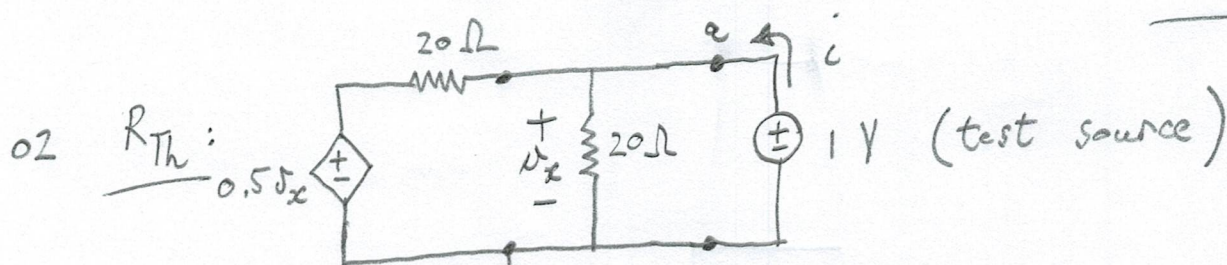
Determine the Thevenin equivalent voltage V_{Th} across terminals $a - b$ and the value of the load resistor R_L connected between terminals $a - b$ for maximum power transfer.

01 V_{Th} . [2]02 R_L . [2]

01 V_{Th} : KCL at top node:

$$\frac{v_x - 0.5v_x}{20} + 3 + \frac{v_x}{20} = 0$$

$$v_x \left(\frac{0.5}{20} + \frac{1}{20} \right) = -3 \therefore v_x = -40 \text{ V} = V_{Th}$$

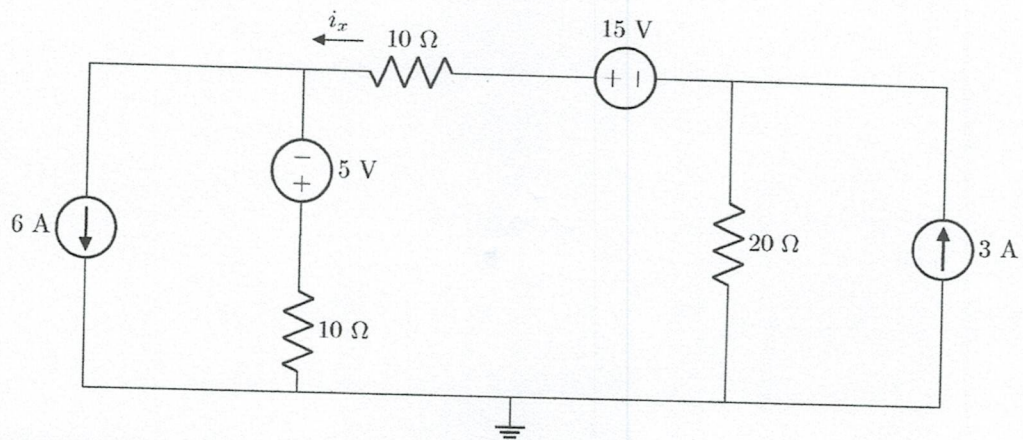


$$\text{KCL: } i = \frac{v_x - 0.5v_x}{20} + \frac{v_x}{20} \text{ but } v_x = 1 \text{ V}$$

$$\therefore i = 0.075 \text{ A}$$

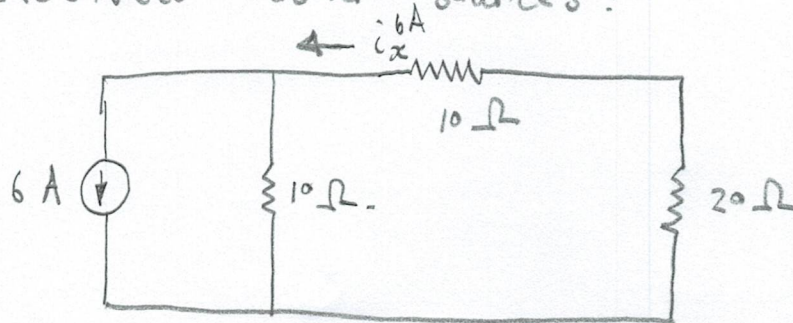
$$\therefore R_{Th} = \frac{1}{i} = 13.33 \Omega = R_L$$

Question 3 [2]



03 Determine the independent contribution of the 6 A current source to i_x . [2]

Deactivate other sources:

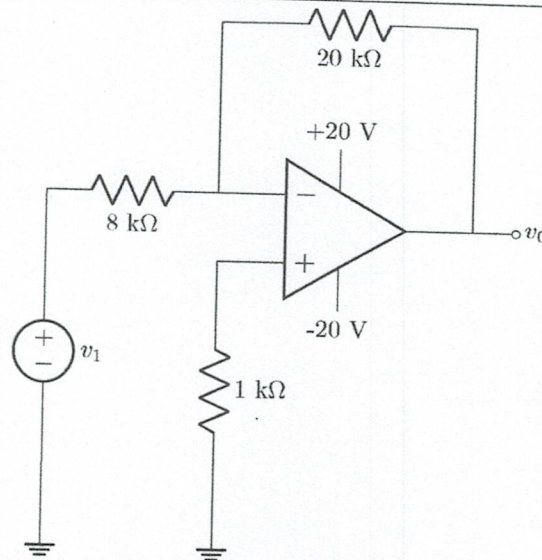


$$i_x = 6 \cdot \frac{10}{10 + (10 + 20)} = 1.5 \text{ A}$$

→

Question 4 [2]

Assume that the op-amp is ideal. Determine the minimum value that v_1 must have for the op-amp to go into negative saturation.

04 v_1 [2]

Inverting amplifier:

$$v_o = -\frac{20000}{8000} v_1$$

During negative saturation, $v_o = -20\text{ V}$

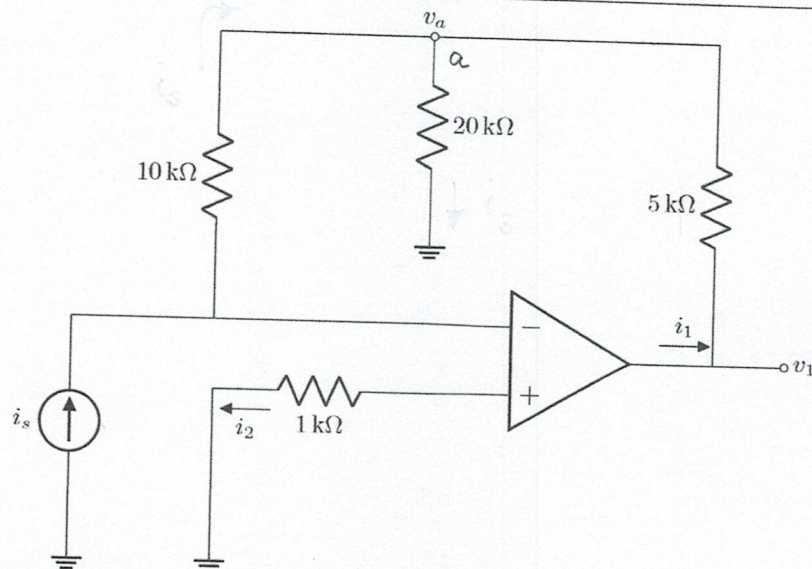
$$-20 = -\frac{5}{2} v_1$$

$$\therefore v_1 = 8\text{ V}$$



Questions 5 to 8 [6]

Assume that the op-amp is ideal, and that $v_a = -10$ V. Find i_s , i_2 , v_1 and i_1 .

05. i_s [1]06. i_2 [1]07. v_1 [2]08. i_1 [2]

$$05. \quad i_s = \frac{0 - v_a}{10\,000} = \frac{0 - (-10)}{10\,000} = 1\text{ mA}$$

$$06. \quad i_2 = 0\text{ A}$$

07. KCL at a:

$$i_s = \frac{v_a}{20\,000} + \frac{v_a - v_1}{5\,000}$$

$$1 \times 10^{-3} = \frac{-10}{20\,000} + \frac{-10}{5\,000} - \frac{v_1}{5\,000} \quad \therefore v_1 = -17.5\text{ V}$$

$$08. \quad i_1 = \frac{v_1 - v_a}{5\,000} = -1.5\text{ mA}$$

07, 08: continuity marking

Question 9 [2]

- 09 Calculate the voltage $v(t)$ at time $t = 2$ s across a capacitor with capacitance $4 \mu\text{F}$ if the current through the capacitor is given by $i(t) = 5 \times 10^{-6}(1 - e^{-3t})$ A, and the initial voltage is $v(0) = 2$ V. [2]

$$v(t) = \frac{1}{C} \int_{t_0}^t i(x) dx + v(t_0)$$

$$v(t=2s) = \frac{1}{4 \times 10^{-6}} \int_0^2 5 \times 10^{-6} (1 - e^{-3x}) dx + 2$$

$$= \frac{5 \times 10^{-6}}{4 \times 10^{-6}} \left[x - \left(-\frac{1}{3}\right) e^{-3x} \right]_0^2 + 2$$

$$= \frac{5 \times 10^{-6}}{4 \times 10^{-6}} \left[2 + \frac{1}{3} e^{-3(2)} - \left(0 + \frac{1}{3}\right) \right] + 2$$

$$= 4 \text{ V}$$

This page is intentionally left open for rough work.

Total Section 1: [16]

Grand Total : [16]